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# Manglares e islas flotantes

## Mangrove Floating Islands: Ecological Functions, Environmental Impacts, Engineering Innovations, and Future Directions

### Quick Reference

#### Key Findings Table

| Theme                   | Key Insights                                                                                       | Supporting Citations    |
|-------------------------|----------------------------------------------------------------------------------------------------|-------------------------|
| Ecological Functions    | Support habitat restoration, biodiversity, nursery grounds, sediment stabilization, and resilience | <div>123456</div>       |
| Environmental Impact    | Improve water quality, enhance nutrient cycling, significant blue carbon sinks                     | <div>78910111213</div>  |
| Microbial Processes     | Mediate nutrient cycling, carbon sequestration, denitrification; vary seasonally/spatially         | <div>141516171819</div> |
| Engineering Innovations | Artificial root systems, modular platforms, hybrid eco-engineering, improved stability             | <div>202122232425</div> |
| Socio-Economic Barriers | High costs, land tenure, governance, need for community-based models                               | <div>24262728</div>     |
| Remote Sensing          | UAV-LiDAR, multispectral, machine learning for monitoring structure and biomass                    | <div>293031</div>       |

### Direct Answer

Research into mangrove floating islands reveals that they serve as multifunctional ecosystems supporting habitat restoration, improving water quality, aiding in carbon sequestration, and stabilizing sediments. Engineering innovations in artificial island design increasingly focus on bio-integrative and modular approaches. Socio-economic challenges, remote sensing for monitoring, and integrated community-based restoration strategies are pivotal areas for future work.

### Study Scope

- Time Period:** Studies span from foundational ecological research to recent engineering and remote sensing advances (last two decades).

- **Disciplines:** Ecology, environmental science, coastal engineering, microbial ecology, socio-economics, remote sensing.
- **Methods:** Field experiments, meta-analyses, numerical modeling, remote sensing (LiDAR, UAV, multispectral), community-based participatory research.

### Assumptions & Limitations

- Most studies focus on tropical and subtropical regions; temperate and arid zone data are less comprehensive.
- Long-term, large-scale field trials of artificial floating islands are limited.
- Socio-economic analyses often lack integration with ecological and engineering data.
- Remote sensing accuracy may vary with sensor type, spatial resolution, and environmental complexity.

### Suggested Further Research

- Quantitative studies on juvenile fish growth/survival on floating islands.
- Long-term monitoring of sediment dynamics and blue carbon sequestration.
- Socio-economic impact assessments and governance model testing.
- Integration of high-resolution remote sensing with community-based management.
- Experimental trials of hybrid restoration models combining technology and local knowledge.

## 1. Introduction

### Overview of Mangrove Floating Islands

Mangrove floating islands are unique coastal features that can be either naturally occurring or artificially engineered. They consist of mangrove vegetation rooted in floating or semi-floating substrates, often forming in estuarine, deltaic, or lagoonal environments. Artificial mangrove floating islands (AMFIs) are increasingly deployed as nature-based solutions for coastal protection, habitat restoration, and climate adaptation, especially in regions facing rapid shoreline erosion and habitat loss. Their global relevance is underscored by their multifunctional ecological roles and their integration into blue carbon and climate resilience strategies [1](#) [20](#).

## 2. Ecological Functions of Mangrove Floating Islands

### Habitat Restoration and Biodiversity Support

Mangrove floating islands are critical for restoring degraded coastal habitats. They:

- Re-establish natural assemblage structures and ecosystem functions, supporting self-sustaining coastal ecosystems [1](#).
- Attenuate wave energy, reduce erosion, and enhance resilience to sea-level rise and storm surges [24](#) [32](#).

- Provide nursery, feeding, and spawning grounds for diverse aquatic fauna (fish, crustaceans, birds), supporting fisheries and biodiversity [33](#) [34](#).
- Serve as stepping stones for macrofauna and microorganisms, enhancing ecosystem rehabilitation [35](#).
- Offer complex vegetation structures and food resources, vital for maintaining coastal biodiversity [36](#) [37](#).
- Enhance local livelihoods and blue carbon sequestration, especially in tropical communities [38](#) [39](#) [40](#).

**Synthesis:** The ecological value of mangrove floating islands lies in their ability to simultaneously restore habitat, support biodiversity, and provide ecosystem services that underpin coastal resilience [1](#) [2](#) [3](#) [4](#) [5](#) [6](#).

### Species-Specific Nursery Functions

- Mangrove species differ in their effectiveness as nursery habitats. For example, *Avicennia marina* thrives in freshwater, while other species may be more salt-tolerant [41](#).
- Nursery value is influenced by root morphology, habitat complexity, and hydrogeomorphological features [3](#) [42](#) [43](#).
- Some fish species are highly dependent on mangroves, while others use alternative habitats, indicating variability in nursery reliance [44](#).
- Mangrove roots provide shelter and food, supporting juvenile development and fisheries productivity [34](#) [45](#).
- There is a research gap in quantifying direct contributions to juvenile growth and survival [46](#).

**Synthesis:** Nursery function is species- and context-dependent, with root morphology and habitat complexity being key determinants of ecological effectiveness [41](#) [42](#) [43](#) [44](#) [47](#).

### Hydrogeomorphological Modulation of Nursery Effectiveness

- Tidal regimes, sediment dynamics, and hydrogeomorphological features (e.g., tidal channels, sediment supply) modulate nursery habitat quality [48](#) [49](#) [50](#) [51](#) [52](#).
- Restored tidal flows enhance sediment supply, propagule dispersal, and mangrove regeneration [53](#) [54](#) [55](#).
- Sediment accretion and surface elevation are critical for resilience to sea-level rise [56](#).
- Propagule dispersal is shaped by hydrodynamics, influencing species zonation and restoration outcomes [57](#).

**Synthesis:** Hydrogeomorphological context is central to the nursery and restoration functions of mangrove floating islands, with tidal connectivity and sediment dynamics underpinning long-term ecological success [48](#) [49](#) [50](#) [51](#) [52](#).

### Long-Term Ecological Impacts of Tidal Restoration

- Reintroduction of tidal flows in degraded systems increases sedimentation, reduces salinity stress, and promotes natural regeneration [53](#) [54](#).
- Tidal restoration leads to rapid ecological recovery, improved seedling survival, and enhanced blue carbon sequestration [58](#).
- Restoration strategies combining hydrologic reconnection, propagule planting, and substrate enhancement accelerate recovery [58](#).

**Synthesis:** Long-term ecological impacts of tidal restoration are overwhelmingly positive, supporting sediment accretion, propagule dispersal, and ecosystem resilience [53](#) [54](#) [55](#) [56](#) [58](#).

### 3. Environmental Impact: Water Quality and Carbon Sequestration

#### Water Quality Improvement and Blue Carbon Storage

- Mangrove floating islands are significant carbon sinks, with stocks of 220–575 Mg C/ha and sequestration potentials up to 990 t CO<sub>2</sub>/ha [7](#) [10](#).
- They improve water quality by stabilizing sediments, enhancing nutrient cycling, and reducing carbon emissions [8](#) [59](#).
- Restoration increases soil carbon and nitrogen sequestration, improving ecosystem health [11](#).
- Efficient water exchange promotes dissolved inorganic carbon outwelling, facilitating long-term sequestration [13](#).
- Mixed-species plantations enhance carbon sequestration and water quality [60](#) [61](#).

**Synthesis:** Mangrove floating islands are integral to blue carbon strategies and water quality improvement, with restoration amplifying these benefits [7](#) [8](#) [9](#) [10](#) [11](#) [12](#) [13](#).

#### Nutrient Cycling and Microbial Community Dynamics

- Mangrove sediments support complex microbial communities that drive nutrient cycling and pollutant removal [14](#) [62](#).
- Tidal inputs and eutrophic conditions shape microbial networks and carbon cycling genes [15](#).
- Microbial mats contribute to nitrogen fixation and denitrification, influencing nutrient availability [19](#).
- Vegetated floating islands enhance water quality via microbial biofilms on roots [63](#).
- Microbial communities are sensitive to land use changes and nutrient loading [64](#) [65](#).

**Synthesis:** Microbial processes are central to the water quality and nutrient cycling functions of mangrove floating islands, with community composition shaped by environmental gradients and anthropogenic impacts [14](#) [15](#) [19](#) [62](#) [63](#) [66](#).

## Seasonal and Spatial Variability in Microbial Functions

- Microbial community composition varies with redox potential, salinity, organic carbon, and season [16](#) [17](#) [67](#).
- Tropical and temperate zones show distinct seasonal patterns in microbial diversity and nutrient cycling [17](#) [68](#).
- Environmental gradients (e.g., pH, oxygen, organic matter) drive spatial heterogeneity in microbial functions [69](#) [70](#).
- Restoration alters microbial community structure, enhancing biogeochemical cycling [71](#).

**Synthesis:** Microbial ecosystem functions on mangrove floating islands are highly dynamic, varying with season, spatial context, and restoration status [16](#) [17](#) [67](#) [68](#) [72](#) [73](#).

## Microbial Enhancement of Carbon Sequestration and Denitrification

- Mangrove-associated microorganisms mediate sulfur, nitrogen, and methane cycles, supporting organic carbon burial and denitrification [18](#) [19](#).
- Artificial floating islands with polyculture planting promote beneficial microbial taxa for nutrient assimilation [74](#).
- Restoration improves soil aggregate stability and microbial network complexity, enhancing carbon sequestration [75](#).
- Microbial decomposition is regulated by phenolic compounds, supporting blue carbon storage [76](#).

**Synthesis:** Microbial communities are pivotal for the carbon and nitrogen cycling functions of both natural and artificial mangrove floating islands, underpinning their role in climate mitigation [18](#) [19](#) [74](#) [75](#) [77](#) [78](#).

## 4. Engineering and Design Innovations in Artificial Mangrove Floating Islands

### Technological Advancements and Design Strategies

- Artificial mangrove root (AMR) systems trap sediment, increase seabed elevation, and facilitate mangrove rehabilitation [20](#).
- Modular construction, adjustable ballast, and integration with hard infrastructure improve stability and adaptability [23](#) [24](#).
- Advanced AFI technologies include temperature control, carbon fiber heating, and big data for environmental adaptation [22](#).
- Hybrid platforms combine landfill and mangrove growth for deep or high-wave coasts [25](#).
- Floating mangrove breakwaters (e.g., *Rhizophora*) can reduce wave energy by up to 80% [32](#).

**Synthesis:** Recent engineering advances focus on hybrid, modular, and bio-integrative designs that enhance ecological function and structural resilience [20](#) [22](#) [23](#) [24](#) [25](#) [32](#) [79](#) [80](#).

### Materials and Structural Stability under High Wave Energy

- Reed-based gabions and organic materials dampen short-period waves, with structure width being a key parameter [21](#).
- Flexible artificial roots mimic natural morphology, enhancing wave damping and reducing forces [81](#).
- Modular design and high porosity improve performance in high-energy environments [23](#).
- Inert plastics and adjustable buoyancy systems extend service life and facilitate maintenance [82](#) [83](#) [84](#).
- Platform size and vegetation porosity are critical for wave attenuation and stability [85](#).

**Synthesis:** Material selection, structural flexibility, and modularity are essential for optimizing artificial mangrove floating islands in dynamic coastal settings [21](#) [23](#) [81](#) [82](#) [83](#) [84](#) [86](#).

### Sediment Accretion and Natural Mangrove Colonization

- AMRs increase seabed elevation by up to 40 cm/year, promoting seedling survival [20](#).
- Root morphology influences sediment trapping; plank roots show highest accretion [87](#).
- Biodegradable frameworks and BESE-Elements aid restoration by trapping sediment and attenuating waves [88](#).
- La regeneración natural y la plantación asistida, combinadas con la restauración del hábitat, mejoran la recuperación. [89](#).

**Síntesis:** Los sistemas de raíces artificiales y los marcos de ingeniería ecológica aceleran la acumulación de sedimentos y facilitan la colonización natural de los manglares, lo que favorece una rápida recuperación del ecosistema. [20](#) [23](#) [81](#) [87](#) [88](#) [90](#).

### Rendimiento hidrodinámico: densidad de la cubierta vegetal y flexibilidad de las raíces.

- La morfología y la densidad del dosel radicular modulan el flujo, la turbulencia y la estabilidad de los sedimentos. [91](#) [92](#) [93](#).
- Las raíces flexibles se deforman bajo las fuerzas hidrodinámicas, aumentando la resistencia y la estabilización del sedimento. [94](#) [95](#).
- Las densas copas de los árboles mejoran la atenuación de las olas; las nuevas fórmulas del coeficiente de arrastre mejoran la precisión de la predicción. [96](#) [97](#).
- La densidad y la morfología de las raíces afectan la retención de sedimentos y la atenuación del flujo, factores críticos para la resiliencia. [98](#) [99](#) [100](#) [101](#).

**Síntesis:** La densidad del dosel y la flexibilidad de las raíces son parámetros de diseño clave para optimizar el rendimiento hidrodinámico y la estabilidad a largo plazo de las islas flotantes artificiales. [91](#) [92](#) [93](#) [94](#) [96](#) [97](#).

## 5. Lagunas en la investigación y direcciones futuras

### Lagunas y prioridades actuales en la investigación

- Datos experimentales limitados sobre el crecimiento/supervivencia de peces juveniles y la dinámica de sedimentos a largo plazo. [24](#) [102](#) [103](#) [104](#) [105](#) [106](#) [107](#).
- Las dimensiones socioeconómicas y las perspectivas de las partes interesadas están insuficientemente representadas.
- Es necesario mejorar la integración de los datos de teledetección, ecológicos y socioeconómicos.
- Existen lagunas de conocimiento sobre los factores biofísicos que afectan a la dispersión de propágulos y a la resiliencia climática.

**Síntesis:** Abordar estas brechas requiere esfuerzos de investigación multidisciplinarios, a largo plazo y que abarquen diversas regiones. [24](#) [102](#) [103](#) [104](#) [105](#) [106](#) [107](#).

### Barreras socioeconómicas y soluciones para el despliegue a gran escala

- Los altos costos, el mantenimiento, la tenencia de la tierra y los problemas de gobernanza dificultan la expansión. [24](#) [26](#) [32](#) [108](#) [109](#) [110](#).
- Los enfoques basados en la comunidad (CBEMR, por sus siglas en inglés), la ingeniería modular y las reformas políticas son soluciones prometedoras.
- La revitalización socioecológica (ecoturismo, educación, medios de vida alternativos) favorece la sostenibilidad.

**Síntesis:** Superar las barreras socioeconómicas requiere estrategias integradas, técnicas, políticas y comunitarias adaptadas a los contextos locales. [24](#) [26](#) [32](#) [108](#) [109](#) [110](#).

### Teledetección para monitoreo y gestión

- Los modelos UAV-LiDAR, multiespectrales y de aprendizaje automático (por ejemplo, XGBoost) proporcionan evaluaciones precisas de biomasa y estructura. [29](#) [30](#) [31](#).
- La fusión de datos multisensor y los algoritmos avanzados mejoran la monitorización en entornos heterogéneos.
- La cartografía a pequeña escala permite una gestión y una planificación de la restauración precisas.

**Síntesis:** Las innovaciones en teledetección están transformando el monitoreo de las islas flotantes de manglares, permitiendo una gestión escalable y de alta resolución. [29](#) [30](#) [103](#) [111](#) [112](#) [113](#) [114](#).

## Marcos de gobernanza y modelos socioeconómicos

- Los marcos de gobernanza inclusivos, basados en el ecosistema y de múltiples niveles son los más eficaces. [26](#) [27](#) [115](#) [116](#) [117](#) [118](#) [119](#).
- La cogestión, los esquemas de pagos por servicios ecosistémicos y la integración con las áreas marinas protegidas facilitan el despliegue a gran escala.
- Los modelos socioeconómicos que integran el ecoturismo, el uso sostenible de los recursos y el empoderamiento de la comunidad mejoran la sostenibilidad.

**Síntesis:** Una gobernanza eficaz requiere coordinación a distintos niveles, coherencia en las políticas y participación activa de la comunidad. [26](#) [27](#) [115](#) [116](#) [117](#) [118](#) [119](#).

## Integración de la restauración comunitaria con la innovación tecnológica.

- CBEMR hace hincapié en las evaluaciones ecológicas e hidrológicas, la participación local y la gestión adaptativa. [120](#) [121](#) [122](#) [123](#).
- La regeneración natural asistida y la ecoingeniería de bajo costo mejoran el éxito de la restauración. [122](#) [124](#) [125](#).
- La combinación de restauración activa y pasiva, desarrollo empresarial y fortalecimiento de las instituciones comunitarias permite alcanzar objetivos tanto ecológicos como de subsistencia. [126](#).
- Las técnicas de restauración innovadoras (por ejemplo, barriles resistentes a la corrosión, diques vivos) pueden superar los desafíos específicos de cada lugar. [127](#) [128](#).

**Síntesis:** La integración de enfoques comunitarios con la innovación tecnológica es esencial para la restauración sostenible y escalable de islas flotantes de manglares. [54](#) [89](#) [120](#) [121](#) [122](#) [123](#) [124](#) [125](#) [127](#) [129](#).

## 6. Implicaciones en el mundo real

- **Protección costera:** Las islas flotantes de manglares ofrecen soluciones basadas en la naturaleza para la estabilización de la línea costera, la atenuación de las olas y la adaptación al cambio climático, reduciendo la dependencia de la infraestructura rígida. [20](#) [24](#) [32](#).
- **Conservación de la biodiversidad:** Apoyan hábitats críticos para la pesca, las aves y otra fauna, lo que sustenta la seguridad alimentaria y la salud de los ecosistemas. [33](#) [34](#).
- **Carbono azul y calidad del agua:** Su papel en el secuestro de carbono y el ciclo de nutrientes los convierte en activos clave en las estrategias de mitigación del cambio climático y mejora de la calidad del agua. [7](#) [10](#) [12](#).
- **Desarrollo socioeconómico:** Los modelos de restauración comunitaria y ecoturismo pueden mejorar los medios de subsistencia, la educación y la gestión local. [118](#) [130](#).

- **Política y gobernanza:** Se necesitan marcos de gobernanza integrados y multinivel para un despliegue sostenible a gran escala. [27](#) [115](#) [116](#).

## 7. Direcciones futuras de la investigación

- **Estudios experimentales:** Cuantificar los impactos directos en el crecimiento, la supervivencia y el movimiento de los peces juveniles; dinámica a largo plazo de los sedimentos y del carbono azul.
- **Integración socioeconómica:** Desarrollar y probar modelos de gobernanza, estrategias de participación de las partes interesadas y evaluaciones de viabilidad económica.
- **Teledetección:** Monitorización avanzada multisensor basada en aprendizaje automático para una gestión adaptativa.
- **Modelos de restauración híbridos:** Combinan la teledetección avanzada con el diseño participativo para obtener soluciones optimizadas y adaptadas al contexto.
- **Adaptación al cambio climático:** Investigar la resiliencia ante el aumento del nivel del mar, los fenómenos meteorológicos extremos y las presiones antropogénicas.

**Síntesis:** Las islas flotantes de manglares están a la vanguardia de la ingeniería ecológica, ofreciendo soluciones escalables y multifuncionales para la resiliencia costera. La integración de la investigación ecológica, de ingeniería y socioeconómica es esencial para aprovechar todo su potencial en la restauración, la mitigación del cambio climático y el desarrollo sostenible. [1](#) [7](#) [20](#) [27](#) [120](#).

## Referencias

1. Mangrove restoration: A potential tool for coastal management in tropical developing countries Kaly, U.L. Ambio, 1998 <https://www.scopus.com/pages/publications/0032465146?origin=scopusAI>
2. An evaluation of the evidence for linkages between mangroves and fisheries: A synthesis of the literature and identification of research directions Manson, F.J., Loneragan, N.R., Skilleter, G.A., Phinn, S.R. Oceanography and Marine Biology, 2005 <https://www.scopus.com/pages/publications/14644418373?origin=scopusAI>
3. Mangroves as nursery habitats for Galapagos reef fishes Aguaiza, C., Mumby, P.J., Rastoin-Laplane, E., (...), Salinas-De-León, P. Marine Ecology Progress Series, 2024 <https://www.scopus.com/pages/publications/85207413888?origin=scopusAI>
4. Resampling 25 years later reveals fewer species but higher abundance of juvenile fishes in a Caribbean mangrove bay Kendall, M.S., Williams, B.L., Ruffo, A., (...), Tobias, W. Bulletin of Marine Science, 2021 <https://www.scopus.com/pages/publications/85101393932?origin=scopusAI>
5. Inferred growth of juvenile French grunts, *Haemulon flavolineatum*, and schoolmaster, *Lutjanus apodus*, in mangrove and seagrass habitats Mateo, I., Durbin, E., Appeldoorn, R., (...), Durant, D. Bulletin of Marine Science, 2011 <https://www.scopus.com/pages/publications/79960480234?origin=scopusAI>
6. Patterns of mangrove productivity and support for marine fauna Ouyang, X., Guo, F. Handbook of Halophytes: From Molecules to Ecosystems towards Biosaline Agriculture, 2021 <https://www.scopus.com/pages/publications/85149352907?origin=scopusAI>

7. Quantifying Carbon Storage and Sequestration in Mangrove Ecosystems of Remote Eastern Indonesian Islands: A Case Study of Akat and Parang Islands Rahman, R., Tuahatu, J.W., Yulianda, F., (...), Widodo, C. *Egyptian Journal of Aquatic Biology and Fisheries*, 2025  
<https://www.scopus.com/pages/publications/105014010357?origin=scopusAI>
8. Carbon stocks in mangrove ecosystems of Sri Lanka: Average contributions and determinants of sequestration potential Perera, K.A.R.S., de Silva, W., Amarasinghe, M.D. *Ocean and Coastal Management*, 2024  
<https://www.scopus.com/pages/publications/85202528517?origin=scopusAI>
9. Carbon sequestration in mangrove ecosystems: Sources, transportation pathways, influencing factors, and its role in the carbon budget Ali, S., Dey, G., Nuong, N.H.K., (...), Chen, C.Y. *Earth-Science Reviews*, 2025  
<https://www.scopus.com/pages/publications/105007761120?origin=scopusAI>
10. Carbon Stock Assessment of Mangrove Vegetation in Anambas Islands Marine Tourism Park, Indonesia Sinaga, R.R.K., Kurniawan, F., Roni, S., (...), Hidayati, J.R. *IOP Conference Series: Earth and Environmental Science*, 2023  
<https://www.scopus.com/pages/publications/85151239590?origin=scopusAI>
11. Potential for carbon and nitrogen sequestration by restoring tidal connectivity and enhancing soil surface elevations in denuded and degraded South Florida mangrove ecosystems Cormier, N., Krauss, K.W., Demopoulos, A.W.J., (...), Flynn, L.L. *Wetland Carbon and Environmental Management*, 2021  
<https://www.scopus.com/pages/publications/85144350251?origin=scopusAI>
12. Contribution of mangroves to coastal carbon cycling in low latitude seas Alongi, D.M., Mukhopadhyay, S.K. *Agricultural and Forest Meteorology*, 2015  
<https://www.scopus.com/pages/publications/84940452526?origin=scopusAI>
13. Insights Into the Comprehensive Carbon Cycle in a Mangrove Ecosystem: A Case Study for Understanding Carbon Burial, Outgassing, and Outwelling in a Subtropical Island Nakamura, W., Watanabe, K., Nakajima, T., (...), Sasaki, J. *Journal of Geophysical Research: Biogeosciences*, 2025  
<https://www.scopus.com/pages/publications/105023707371?origin=scopusAI>
14. Enhanced sediment microbial diversity in mangrove forests: Indicators of nutrient status in coastal ecosystems Li, B., Xia, Y., Chen, X., (...), Ren, H. *Marine Pollution Bulletin*, 2025  
<https://www.scopus.com/pages/publications/85211316919?origin=scopusAI>
15. The influence of tide-brought nutrients on microbial carbon metabolic profiles of mangrove sediments Li, X., Cheng, X., Cheng, K., (...), Zhou, J. *Science of the Total Environment*, 2024  
<https://www.scopus.com/pages/publications/85174333995?origin=scopusAI>
16. Redox potential drives divergent microbial carbon fixation in mangrove wetland sediments, with ammonium exerting context-dependent effects Yan, S., Li, R., Shen, X., (...), Xie, S. *Marine Pollution Bulletin*, 2026  
<https://www.scopus.com/pages/publications/105030921082?origin=scopusAI>
17. Spatial and temporal heterogeneity in the structure and function of sediment bacterial communities of a tropical mangrove forest Behera, P., Mohapatra, M., Kim, J.Y., (...), Rastogi, G. *Environmental Science and Pollution Research*, 2019  
<https://www.scopus.com/pages/publications/85058451822?origin=scopusAI>
18. Mangrove microbiome reveals importance of sulfur metabolism in tropical coastal waters Nóbrega, M.S., Silva, B.S., Tschoeke, D.A., (...), Thompson, F.L. *Science of the Total Environment*, 2022  
<https://www.scopus.com/pages/publications/85120404862?origin=scopusAI>
19. Seasonal patterns of nitrogen fixation and denitrification in oceanic mangrove habitats Lee, R.Y., Joye, S.B. *Marine Ecology Progress Series*, 2006  
<https://www.scopus.com/pages/publications/33644515359?origin=scopusAI>
20. A bioinspired hybrid nature-based solution using artificial mangrove roots to enhance coastal resilience and mangrove rehabilitation Charoenlerkthawin, W., Burnett, W.C., Sattabongkot, T., (...), Bidorn, B. *Journal of Environmental Management*, 2026  
<https://www.scopus.com/pages/publications/105024312822?origin=scopusAI>

21. Large-scale investigation of wave dampening characteristics of organic, artificial floating islands Landmann, J., Hammer, T.C., Günther, H., Hildebrandt, A. Ecological Engineering, 2022 <https://www.scopus.com/pages/publications/85130787327?origin=scopusAI>
22. A new mechanical ecological floating island based on refrigerant and big data Zou, Z. 2022 IEEE 2nd International Conference on Electronic Technology, Communication and Information, ICETCI 2022, 2022 <https://www.scopus.com/pages/publications/85136331058?origin=scopusAI>
23. Field Evidence of a Nature-Inspired Solution to Mitigate the Impacts of Hard Coastal Structures Charoenlerkthawin, W., Sattabongkot, T., Srinil, N., (...), Bidorn, B. Ocean-Land-Atmosphere Research, 2025 <https://www.scopus.com/pages/publications/105021426722?origin=scopusAI>
24. Overcoming Engineering Challenges in Floating Mangrove Wave Attenuator Design Kathekeyan, M. Australasian Coasts and Ports 2025 Conference, 2025 <https://www.scopus.com/pages/publications/105031753172?origin=scopusAI>
25. "Adapted mangrove on hybrid platform" – Coupling of ecological and engineering principles against coastal hazards Takagi, H. Results in Engineering, 2019 <https://www.scopus.com/pages/publications/85075799127?origin=scopusAI>
26. Policy challenges and approaches for the conservation of mangrove forests in Southeast Asia Friess, D.A., Thompson, B.S., Brown, B., (...), Sidik, F. Conservation biology : the journal of the Society for Conservation Biology, 2016 <https://www.scopus.com/pages/publications/84985869862?origin=scopusAI>
27. Local-scale impacts of mangrove restoration and conservation on coastal communities in two Vietnamese deltas: Socio-economic and institutional dynamics Ebeler, L., Balke, T., Laurie, E., (...), Renaud, F.G. Ocean and Coastal Management, 2025 <https://www.scopus.com/pages/publications/105008206960?origin=scopusAI>
28. A warning about the double-edged sword of integrated aquaculture: Mangrove degradation ongoing in Bac Lieu, Vietnam, as a case study Van Viet, T., Tsai, W.-P., Huynh, H.H. Aquaculture, 2026 <https://www.scopus.com/pages/publications/105027997523?origin=scopusAI>
29. Aboveground mangrove biomass estimation in Beibu Gulf using machine learning and UAV remote sensing Tian, Y., Huang, H., Zhou, G., (...), Lin, J. Science of the Total Environment, 2021 <https://www.scopus.com/pages/publications/85103954946?origin=scopusAI>
30. Aboveground Carbon Estimation in a Mangrove Ecosystem Using UAV-Based Remote Sensing and Machine Learning Duan, M., Sanchez-Azofeifa, A., Abdulmajeed, M., (...), Kasraee, N. Ecological Indicators, 2025 <https://www.scopus.com/pages/publications/105012246959?origin=scopusAI>
31. RAP-CNN: A Deep Learning Framework for Mangrove Aboveground Biomass Estimation Using Multisource Remote Sensing Data Zhuang, J., Huang, P., Liu, G., (...), Chen, H. IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing, 2026 <https://www.scopus.com/pages/publications/105031528673?origin=scopusAI>
32. Are floating mangrove breakwaters effective for wave attenuation? - A Literature Review Risheharan, K., Baumeister, J., Adame, M.F. Ecological Engineering, 2025 <https://www.scopus.com/pages/publications/85209911819?origin=scopusAI>
33. Mangrove governance: Establish a new paradigm of mangrove management "from village to the world" Yonvitner, Ferdinan, Agus, S.B., (...), Akmal, S.G. IOP Conference Series: Earth and Environmental Science, 2020 <https://www.scopus.com/pages/publications/85094185879?origin=scopusAI>
34. The habitat functions of mangroves for aquatic fauna and fisheries Xu, S., Chen, Z., Li, S. Shengtai Xuebao/Acta Ecologica Sinica, 2010 <https://www.scopus.com/pages/publications/77949764251?origin=scopusAI>
35. Biodiversity Associated to Floating Wetland Islands Calheiros, C.S.C., Ilarri, M., Pereira, S.I.A. Environmental Science and Engineering, 2023 <https://www.scopus.com/pages/publications/85164266522?origin=scopusAI>

36. Mangrove fauna of Asia Rajpar, M.N., Zakaria, M. Mangrove Ecosystems of Asia: Status, Challenges and Management Strategies, 2014 <https://www.scopus.com/pages/publications/84929650228?origin=scopusAI>
37. Assessing the fauna diversity of Marudu Bay mangrove forest, Sabah, Malaysia, for future conservation Zakaria, M., Rajpar, M.N. Diversity, 2015 <https://www.scopus.com/pages/publications/84934763810?origin=scopusAI>
38. Mangroves Sustaining Biodiversity, Local Livelihoods, Blue Carbon, and Local Resilience in Verde Island Passage in Luzon, Philippines Gevaña, D.T., Villanueva, C.M.M., Garcia, J.E., Camacho, L.D. Mangroves: Biodiversity, Livelihoods and Conservation, 2022 <https://www.scopus.com/pages/publications/85160793225?origin=scopusAI>
39. Training and Managing Local Coastal Mangrove Dwellers for Climate Change Solutions Masagca, J.T., Morales, M.I., Masagca, M.T. Handbook of Climate Change Management: Research, Leadership, Transformation, 2021 <https://www.scopus.com/pages/publications/85161954460?origin=scopusAI>
40. Mangrove Forest Ecosystem: Services, Conservation, Restoration and Carbon Finance Chanda, S. Forests and Climate Change: Biological Perspectives on Impact, Adaptation, and Mitigation Strategies, 2024 <https://www.scopus.com/pages/publications/105005663406?origin=scopusAI>
41. GROWTH RATE AND SURVIVORSHIP OF RHIZOPHORA MUCRONATA, AVICENNIA MARINA, AND CERIOPS TAGAL SEEDLINGS WITH FRESHWATER AND SEAWATER TREATMENT FOR MANGROVE PROPAGATION IN NURSERIES Etongo, D., D'offay, K., Vel, T., (...), Henriette, E. Applied Ecology and Environmental Research, 2022 <https://www.scopus.com/pages/publications/85143393163?origin=scopusAI>
42. Exploration of data on the formation of two mangrove seedling species to establish a growing stability point Bimantara, Y., Basyuni, M., Selamat, B. IOP Conference Series: Earth and Environmental Science, 2021 <https://www.scopus.com/pages/publications/85104872622?origin=scopusAI>
43. Impacts of different salinity levels on seedling growth and survival of black mangrove (Avicennia germinans) Matto, A.A., Jaikishun, S., Ram, M. Asian Journal of Forestry, 2023 <https://www.scopus.com/pages/publications/85189354899?origin=scopusAI>
44. Shallow patch reefs as alternative habitats for early juveniles of some mangrove/seagrass-associated fish species in Bermuda Huijbers, C.M., Grol, M.G.G., Nagelkerken, I. Revista de Biología Tropical, 2008 <https://www.scopus.com/pages/publications/70149090175?origin=scopusAI>
45. The habitat function of mangroves for terrestrial and marine fauna: A review Nagelkerken, I., Blaber, S.J.M., Bouillon, S., (...), Somerfield, P.J. Aquatic Botany, 2008 <https://www.scopus.com/pages/publications/45049083887?origin=scopusAI>
46. Are mangroves nursery habitat for transient fishes and decapods? Sheridan, P., Hays, C. Wetlands, 2003 <https://www.scopus.com/pages/publications/0042069787?origin=scopusAI>
47. Effects of hydrographic conditions of ponds on juvenile fish assemblages in the Kakum mangrove system, Ghana Levy, E.S., Asare, N.K., Yankson, K., Wubah, D.A. Regional Studies in Marine Science, 2015 <https://www.scopus.com/pages/publications/84939817641?origin=scopusAI>
48. Nature-based solutions on floodplain restoration with coupled propagule dispersal simulation and stepping-stone approach to predict mangrove encroachment in an estuary Shih, S.-S., Huang, Z.-Z., Hsu, Y.-W. Science of the Total Environment, 2022 <https://www.scopus.com/pages/publications/85136246343?origin=scopusAI>
49. Tidal-scale flow routing and sedimentation in mangrove forests: Combining field data and numerical modelling Horstman, E.M., Dohmen-Janssen, C.M., Bouma, T.J., Hulscher, S.J.M.H. Geomorphology, 2015 <https://www.scopus.com/pages/publications/84907830992?origin=scopusAI>
50. On the ecogeomorphological feedbacks that control tidal channel network evolution in a sandy mangrove setting Van Maanen, B., Coco, G., Bryan, K.R. Proceedings of the Royal Society A: Mathematical, Physical and

Engineering Sciences, 2015 <https://www.scopus.com/pages/publications/84939857039?origin=scopusAI>

51. Tidal, diel and seasonal effects on intertidal mangrove fish in a high-rainfall area of the Tropical Eastern Pacific Castellanos-Galindo, G.A., Krumme, U. Marine Ecology Progress Series, 2013 <https://www.scopus.com/pages/publications/84889669168?origin=scopusAI>

52. Demersal fish and shrimp abundance in relation to mangrove hydrogeomorphological metrics Jamizan, A.R., Chong, V.C. Sains Malaysiana, 2017 <https://www.scopus.com/pages/publications/85015763786?origin=scopusAI>

53. Effectiveness of hydrological restoration in the mangrove of a coastal lagoon in the Gulf of Mexico López-Adame, H., Zaragoza-Méndez, F.Á., Lara-Domínguez, A.L., (...), López-Portillo, J. Ecological Engineering, 2026 <https://www.scopus.com/pages/publications/105024350604?origin=scopusAI>

54. Managing sediment dynamics through reintroduction of tidal flow for mangrove restoration in abandoned aquaculture ponds Sidik, F., Kusuma, D.W., Priyono, B., (...), Lovelock, C.E. Dynamic Sedimentary Environments of Mangrove Coasts, 2020 <https://www.scopus.com/pages/publications/85108378871?origin=scopusAI>

55. Modelling drivers of mangrove propagule dispersal and restoration of abandoned shrimp farms Di Nitto, D., Erftemeijer, P.L.A., Van Beek, J.K.L., (...), Koedam, N. Biogeosciences, 2013 <https://www.scopus.com/pages/publications/84879083731?origin=scopusAI>

56. Recent tidal-flat evolution and mangrove-habitat expansion: Application of radioisotope dating to environmental reconstruction Swales, A., Bentley, S.J. IAHS-AISH Publication, 2008 <https://www.scopus.com/pages/publications/62949196367?origin=scopusAI>

57. Propagule dispersal determines mangrove zonation at intertidal and Estuarine scales Wang, W., Li, X., Wang, M. Forests, 2019 <https://www.scopus.com/pages/publications/85063877950?origin=scopusAI>

58. A multiple habitat restoration strategy in a semi-enclosed Florida embayment, combining hydrologic restoration, mangrove propagule plantings and oyster substrate additions Milbrandt, E.C., Thompson, M., Coen, L.D., (...), Ward, K. Ecological Engineering, 2015 <https://www.scopus.com/pages/publications/84938065528?origin=scopusAI>

59. Nutrient Cycles, Protection and Management of Mangroves Ecosystem: A Sustainable Approach Singh, N., Yadav, N., Srivastava, S. Green Energy and Technology, 2025 <https://www.scopus.com/pages/publications/105008269145?origin=scopusAI>

60. Cost-benefit analysis of 'Blue Carbon' sequestration by plantation of few key mangrove species at Sundarban Biosphere Reserve, India Chowdhury, A., Naz, A., Bhattacharyya, S., Sanyal, P. Carbon Management, 2018 <https://www.scopus.com/pages/publications/85059889507?origin=scopusAI>

61. Carbon payments for mangrove conservation: Ecosystem constraints and uncertainties of sequestration potential Alongi, D.M. Environmental Science and Policy, 2011 <https://www.scopus.com/pages/publications/79955728811?origin=scopusAI>

62. Detoxification efficacy of heavy metal contaminants using mangrove-associated microorganisms Jeyanthi, V., Govindarajan, V.K. Mangrove Microbiome: Diversity and Bioprospecting, 2025 <https://www.scopus.com/pages/publications/105005567527?origin=scopusAI>

63. The ability of vegetated floating Islands to improve water quality in natural and constructed wetlands: A review Masters, B. Water Practice and Technology, 2012 <https://www.scopus.com/pages/publications/84857559363?origin=scopusAI>

64. Nitrogen addition alters composition, diversity, and functioning of microbial communities in mangrove soils: An incubation experiment Craig, H., Antwis, R.E., Cordero, I., (...), Simpson, L.T. Soil Biology and Biochemistry, 2021 <https://www.scopus.com/pages/publications/85097041446?origin=scopusAI>

65. Sensitivity of the mangrove-estuarine microbial community to aquaculture effluent Erazo, N.G., Bowman, J.S. *iScience*, 2021 <https://www.scopus.com/pages/publications/85101875633?origin=scopusAI>
66. Effects of tidal influence on the structure and function of prokaryotic communities in the sediments of a pristine Brazilian mangrove De Santana, C.O., Spealman, P., Melo, V.M.M., (...), Chinalia, F.A. *Biogeosciences*, 2021 <https://www.scopus.com/pages/publications/85103766480?origin=scopusAI>
67. Seasonal dynamics of bacterial communities in mangrove sediments of Shupaisha island, Zhejiang Province, China He, M., Jiang, S., Li, X., (...), Shui, B. *Frontiers in Microbiology*, 2025 <https://www.scopus.com/pages/publications/105000016387?origin=scopusAI>
68. Seasonal Fluctuations of Nutrient Characteristics and Interfacial Fluxes in Mangrove Intertidal Groundwater Liu, Z., Li, G., Li, Z., (...), Xiao, K. *Journal of Geophysical Research: Oceans*, 2026 <https://www.scopus.com/pages/publications/105032830775?origin=scopusAI>
69. Vertical and spatial variations of microbial communities in sediment cores from the mangrove of Gaoqiao National Nature Reserve Huang, J., Tan, Z., Sun, R., (...), Li, C. *Marine Pollution Bulletin*, 2026 <https://www.scopus.com/pages/publications/105029964886?origin=scopusAI>
70. Effects of Environmental and Spatial Variables on Bacteria in Zhanjiang Mangrove Sediments Zhu, D.-H., Song, Q.-L., Nie, F.-H., (...), Chen, J.-J. *Current Microbiology*, 2022 <https://www.scopus.com/pages/publications/85124577329?origin=scopusAI>
71. Response of sediment microbial community composition and function to mangrove restoration from an aquaculture pond in Southern China Shi, R., Han, T., Zhang, H., (...), Qi, Z. *Environmental Research*, 2026 <https://www.scopus.com/pages/publications/105027092539?origin=scopusAI>
72. Bacterial community structure of mangrove sediments in relation to environmental variables accessed by 16S rRNA gene-denaturing gradient gel electrophoresis fingerprinting Zhang, Y., Dong, J., Yang, B., (...), Zhang, S. *Scientia Marina*, 2009 <https://www.scopus.com/pages/publications/70149083244?origin=scopusAI>
73. The Right Place at the Right Time: Seasonal Variation of Bacterial Communities in Arid *Avicennia marina* Soils in the Red Sea Is Specific to Its Position in the Intertidal Thomson, T., Ellis, J.I., Fusi, M., (...), Jones, B.H. *Frontiers in Ecology and Evolution*, 2022 <https://www.scopus.com/pages/publications/85128614585?origin=scopusAI>
74. Characterization of microbial communities in the rhizosphere and water of a field-scale artificial floating island system for nutrient removal Chen, Z., Li, L., Kim, M., (...), Costa, O.S. *Environmental Pollution*, 2025 <https://www.scopus.com/pages/publications/105013579733?origin=scopusAI>
75. Mangrove wetland recovery enhances soil carbon sequestration capacity of soil aggregates and microbial network stability in southeastern China Hou, N., Yang, X., Wang, W., (...), Peñuelas, J. *Science of the Total Environment*, 2024 <https://www.scopus.com/pages/publications/85202206625?origin=scopusAI>
76. Decomposition as a regulator of carbon accretion in mangroves: a review Friesen, S.D., Dunn, C., Freeman, C. *Ecological Engineering*, 2018 <https://www.scopus.com/pages/publications/85026637376?origin=scopusAI>
77. Microbial Community Structure of the Sundarbans Mangrove Ecosystem Pramanik, A., Bhattacharyya, M. *Microbial Diversity in the Genomic Era: Functional Diversity and Community Analysis, Second Edition*, 2024 <https://www.scopus.com/pages/publications/105001252735?origin=scopusAI>
78. An introduction to mangrove ecosystem and their associated microorganisms Nagarajan, T., Veilumuthu, P., Srinithan, T., Christopher, J.G. *Mangrove Microbiome: Diversity and Bioprospecting*, 2025 <https://www.scopus.com/pages/publications/105005569594?origin=scopusAI>
79. Coastal Protection Using Integration of Mangroves with Floating Barges: An Innovative Concept Raju, R.D., Arockiasamy, M. *Journal of Marine Science and Engineering*, 2022 <https://www.scopus.com/pages/publications/85132820144?origin=scopusAI>

80. Technologies for performance intensification of floating treatment wetland – An explicit and comprehensive review Zhang, F., Wang, J., Li, L., (...), Zhao, Y. *Chemosphere*, 2024  
<https://www.scopus.com/pages/publications/85178006363?origin=scopusAI>
81. Numerical Investigation of Wave-Induced Damping and Dynamic Response of Artificial Mangrove Roots Aye, Y.N., Srinil, N., Bidorn, B., (...), Charoenlerkthawin, W. *Proceedings of the International Conference on Offshore Mechanics and Arctic Engineering - OMAE*, 2025  
<https://www.scopus.com/pages/publications/105015135189?origin=scopusAI>
82. Large scale floating wetland application for national aquarium McCarty, M., Ceci, J., Streb, C. *Australasian Coasts and Ports 2017 Conference*, 2017 <https://www.scopus.com/pages/publications/85055087000?origin=scopusAI>
83. A case study and recommendation for large scale floating wetlands McCarty, M., Ceci, J., Streb, C. *International Conference on Sustainable Infrastructure 2017: Technology - Proceedings of the International Conference on Sustainable Infrastructure 2017*, 2017  
<https://www.scopus.com/pages/publications/85035102479?origin=scopusAI>
84. Ports 2019 Large Scale Floating Wetlands for Urban Waterfronts McCarty, M., Ravestein, A., Streb, C., O'Heney, D. *Ports 2019: Port Planning and Development - Papers from Sessions of the 15th Triennial International Conference*, 2019 <https://www.scopus.com/pages/publications/85072913136?origin=scopusAI>
85. Hydrodynamic characteristics of a floating vegetated platform exposed to regular waves Li, J., Yin, Z., Wang, Y. *Physics of Fluids*, 2025 <https://www.scopus.com/pages/publications/85218343510?origin=scopusAI>
86. Wave attenuation and stability of floating wetlands in an erosional coast Gong, Z., Zhou, X., Chen, X., (...), Zhang, Q. *Hydro-Science and Engineering*, 2025 <https://www.scopus.com/pages/publications/105032779636?origin=scopusAI>
87. Differences in the vertical accretion of sediment among mangrove species with different aerial root types Du, Q., Qin, Z., Ming, S., Zhang, C. *Estuarine, Coastal and Shelf Science*, 2021  
<https://www.scopus.com/pages/publications/85105270726?origin=scopusAI>
88. Eco-friendly structures for sustainable mangrove restoration Mohan, M., Selvam, P.P., Ewane, E.B., (...), Burt, J.A. *Science of the Total Environment*, 2025 <https://www.scopus.com/pages/publications/105002798099?origin=scopusAI>
89. To Plant or Not to Plant: When can Planting Facilitate Mangrove Restoration? van Bijsterveldt, C.E.J., Debrot, A.O., Bouma, T.J., (...), van Wesenbeeck, B.K. *Frontiers in Environmental Science*, 2022  
<https://www.scopus.com/pages/publications/85125092831?origin=scopusAI>
90. Vegetation-induced soil stabilization in coastal area: An example from a natural mangrove forest Karimi, Z., Abdi, E., Deljouei, A., (...), Hales, T.C. *Catena*, 2022  
<https://www.scopus.com/pages/publications/85131129397?origin=scopusAI>
91. Root Canopy Morphology Controls on Hydrodynamics and Turbulence Around Floating Vegetation Islands Anjum, N., Iqbal, S., Ashraf, J. *Ecohydrology*, 2026  
<https://www.scopus.com/pages/publications/105026856363?origin=scopusAI>
92. Numerical Study of the Hydrodynamics of Mangrove-Inspired Structures for Coastal Protection Quinones, J.J., Esquivel-Puentes, H.A., Nelson, J.O., (...), Castillo, L. *Oceans Conference Record (IEEE)*, 2022  
<https://www.scopus.com/pages/publications/85145770869?origin=scopusAI>
93. Hydrodynamics of mangrove-type root models: The effect of porosity, spacing ratio and flexibility Kazemi, A., Van De Riet, K., Curet, O.M. *Bioinspiration and Biomimetics*, 2017  
<https://www.scopus.com/pages/publications/85033390289?origin=scopusAI>
94. The Deformation and Shear Vortex Width of Flexible Vegetation Roots in an Artificial Floating Bed Channel Qi, Y., Bai, Y., Cao, X., Li, E. *Sustainability (Switzerland)*, 2022

<https://www.scopus.com/pages/publications/85138949194?origin=scopusAI>

95. Drag coefficient and flow structure downstream of mangrove root-type models through PIV and direct force measurements Kazemi, A., Van De Riet, K., Curet, O.M. *Physical Review Fluids*, 2018  
<https://www.scopus.com/pages/publications/85051138317?origin=scopusAI>
96. Experimental investigation of wave attenuation by mangrove forests with submerged canopies Zhang, R., Chen, Y., Lei, J., (...), Stive, M.J.F. *Coastal Engineering*, 2023  
<https://www.scopus.com/pages/publications/85172185354?origin=scopusAI>
97. Wave attenuation by juvenile and mature mangrove *Kandelia Obovata* with flexible canopies Hang, C., Pan, J., Ju, L., (...), Xie, D. *Applied Ocean Research*, 2025  
<https://www.scopus.com/pages/publications/85216545722?origin=scopusAI>
98. Differential rates of vertical accretion and elevation change among aerial root types in Micronesian mangrove forests Krauss, K.W., Allen, J.A., Cahoon, D.R. *Estuarine, Coastal and Shelf Science*, 2003  
<https://www.scopus.com/pages/publications/0037307601?origin=scopusAI>
99. Sensitivity of the sediment trapping capacity of an estuarine mangrove forest Willemsen, P.W.J.M., Horstman, E.M., Borsje, B.W., (...), Dohmen-Janssen, C.M. *Geomorphology*, 2016  
<https://www.scopus.com/pages/publications/84982293422?origin=scopusAI>
100. Hydro-sedimentary processes of mangrove wetland over a meso-macrotidal estuary Zeng, W., Dai, Z., Luo, J., (...), Wang, R. *Marine Geology*, 2026 <https://www.scopus.com/pages/publications/105022109759?origin=scopusAI>
101. Attenuation of Hydro-Sediment Dynamics Progradation Over Mangrove Wetland Jiang, W., Huang, Z., Dai, Z., (...), Liang, X. *Journal of Geophysical Research: Oceans*, 2025  
<https://www.scopus.com/pages/publications/85219078062?origin=scopusAI>
102. Knowledge shortfalls and research priorities for Philippine mangroves in the fast-changing world Agduma, A.R., Tanalgo, K.C., Millondaga, A.M., (...), Cao, K.-F. *Ocean and Coastal Management*, 2024  
<https://www.scopus.com/pages/publications/85196059933?origin=scopusAI>
103. Surface area to volume scaling of alpha shapes and convex hulls fit to lidar derived mangrove island point clouds Price, C.A., Papeş, M., Sgouralis, I., Schroeder, T.A. *Remote Sensing Letters*, 2024  
<https://www.scopus.com/pages/publications/85210590248?origin=scopusAI>
104. Exploring Global Research Trends in Mangrove Ecosystems: A Bibliometric Analysis of Key Issues and Future Directions (1996–2022) Qin, Y., Zhu, G., Liang, M., Feng, B. *Land Degradation and Development*, 2025  
<https://www.scopus.com/pages/publications/86000262169?origin=scopusAI>
105. Mapping Knowledge and Stakeholder Engagement in Mangrove Ecosystem Service Valuation: Insights from a Bibliometric Analysis of the Caribbean and the Gulf of Mexico Kelly-Fair, M., Lippmann, S., Snow, E., (...), Gopal, S. *Journal of Marine Science and Engineering*, 2026  
<https://www.scopus.com/pages/publications/105030157242?origin=scopusAI>
106. Global-scale dispersal and connectivity in mangroves Van der Stocken, T., Carroll, D., Menemenlis, D., (...), Koedam, N. *Proceedings of the National Academy of Sciences of the United States of America*, 2019  
<https://www.scopus.com/pages/publications/85060049994?origin=scopusAI>
107. Remote sensing of mangrove ecosystems: A review Kuenzer, C., Bluemel, A., Gebhardt, S., (...), Dech, S. *Remote Sensing*, 2011 <https://www.scopus.com/pages/publications/80052503089?origin=scopusAI>
108. Rates and drivers of mangrove deforestation in Southeast Asia, 2000-2012 Richards, D.R., Friess, D.A. *Proceedings of the National Academy of Sciences of the United States of America*, 2016  
<https://www.scopus.com/pages/publications/84954516431?origin=scopusAI>

109. CASE STUDY: Community based ecological mangrove rehabilitation (CBEMR) in Indonesia from small (12-33 ha) to medium scales (400 ha) with pathways for adoption at larger scales (> 5000 ha) Brown, B., Fadillah, R., Nurdin, Y., (...), Ahmad, R. Sapiens, 2014  
<https://www.scopus.com/pages/publications/84901763479?origin=scopusAI>
110. Aquaculture induced erosion of tropical coastlines throws coastal communities back into poverty van Wesenbeeck, B.K., Balke, T., van Eijk, P., (...), Winterwerp, J.C. Ocean and Coastal Management, 2015  
<https://www.scopus.com/pages/publications/84942802062?origin=scopusAI>
111. Classification of Mangrove Vegetation Structure using Airborne LiDAR in Ratai Bay, Lampung Province, Indonesia Wijaya, M.S., Kamal, M., Widayani, P., Arjasakusuma, S. Geoplanning, 2023  
<https://www.scopus.com/pages/publications/85182380833?origin=scopusAI>
112. Classification of mangrove vegetation structures using GEOBIA based on Pleiades-1 imagery and airborne LiDAR Wijaya, M.S., Kamal, M., Widayani, P. Regional Studies in Marine Science, 2025  
<https://www.scopus.com/pages/publications/105022101395?origin=scopusAI>
113. Aboveground biomass of typical invasive mangroves and its distribution patterns using UAV-LiDAR data in a subtropical estuary: Maoling River estuary, Guangxi, China Tian, Y., Zhang, Q., Huang, H., (...), Lin, J. Ecological Indicators, 2022 <https://www.scopus.com/pages/publications/85124809407?origin=scopusAI>
114. Coupling UAV Hyperspectral and LiDAR Data for Mangrove Classification Using XGBoost in China's Pinglu Canal Estuary Ou, J., Tian, Y., Zhang, Q., (...), Lin, J. Forests, 2023  
<https://www.scopus.com/pages/publications/85172722682?origin=scopusAI>
115. Governance analysis of a community managed small-scale crab fishery in Madagascar: novel use of an empirical framework Long, S., Jones, P.J.S., Randriana, Z., Hadj-Hammou, J. Marine Policy, 2021  
<https://www.scopus.com/pages/publications/85035222651?origin=scopusAI>
116. Governance and implementation challenges for mangrove forest Payments for Ecosystem Services (PES): Empirical evidence from the Philippines Thompson, B.S., Primavera, J.H., Friess, D.A. Ecosystem Services, 2017 <https://www.scopus.com/pages/publications/85008642064?origin=scopusAI>
117. Governance and the mangrove commons: Advancing the cross-scale, nested framework for the global conservation and wise use of mangroves Walker, J.E., Ankersen, T., Barchiesi, S., (...), Angelini, C. Journal of Environmental Management, 2022 <https://www.scopus.com/pages/publications/85126610791?origin=scopusAI>
118. Formulating a revitalization strategy for a sustainable mangrove environment with natural and human resource management Suswadi, Irawan, N.C., Esthi, R.B., Sari, E.N. IOP Conference Series: Earth and Environmental Science, 2023 <https://www.scopus.com/pages/publications/85179824147?origin=scopusAI>
119. Blue carbon management integrating socioeconomic and environmental interconnectivity in Southeast Asia: an urgent climate priority Becker, A., Bercovici, S.K., Choo, J., (...), Evans, C. Frontiers in Marine Science, 2025 <https://www.scopus.com/pages/publications/105017095454?origin=scopusAI>
120. Stakeholder perceptions on Community-Based Ecological Mangrove Restoration (CBEMR): a case study in Thailand Lhosupasirirat, P., Dahdouh-Guebas, F., Hugé, J., (...), Enright, J. Restoration Ecology, 2023  
<https://www.scopus.com/pages/publications/85150922233?origin=scopusAI>
121. Effectiveness of community-based mangrove management for biodiversity conservation: A case study from Central Java, Indonesia Damastuti, E., de Groot, R., Debrot, A.O., Silvius, M.J. Trees, Forests and People, 2022 <https://www.scopus.com/pages/publications/85123943882?origin=scopusAI>
122. Post-Rehabilitation Dynamics of Mangrove Vegetation in Tarakan, North Kalimantan Wiharyanto, D., Ilman, M., Salim, G., (...), Mujiyanto, M. Ilmu Kelautan: Indonesian Journal of Marine Sciences, 2025  
<https://www.scopus.com/pages/publications/105018015049?origin=scopusAI>
123. Mangrove Rehabilitation and Restoration as Experimental Adaptive Management Ellison, A.M., Felson, A.J., Friess, D.A. Frontiers in Marine Science, 2020 <https://www.scopus.com/pages/publications/85085772194?origin=scopusAI>

origin=scopusAI

124. Environmental Education Model in Mangrove Ecosystem Service Enhancement Alifi, G.J., Dewi, A., Iswandi, I. E3S Web of Conferences, 2025 [https://www.scopus.com/pages/publications/105031266307?](https://www.scopus.com/pages/publications/105031266307?origin=scopusAI)  
origin=scopusAI

125. Community based ecological mangrove rehabilitation (CBEMR) in Indonesia: From small (12-33 ha) to medium scales (400 ha) with pathways for adoption at larger scales (>5000 ha) Brown, B., Fadillah, R., Nurdin, Y., (...), Ahmad, R. Sapiens, 2014 <https://www.scopus.com/pages/publications/84902522965?origin=scopusAI>

126. Systematic review on the implementation of mangrove community-based restoration in Indonesia and beyond Kusumadewi, S.D., Purnomo, H., Nadhira, S., Putrizulfan, R. IOP Conference Series: Earth and Environmental Science, 2024 <https://www.scopus.com/pages/publications/85189366889?origin=scopusAI>

127. Utilization of a new restoration technique for the rehabilitation of a degraded mangrove ecosystem: a case study from Koggala Lagoon, Sri Lanka Dhanushka, M.G.G., Prasanna, M.G.M., Jayasuriya, K.M.G.G., Vitanage, I.H. Journal of Threatened Taxa, 2023 [https://www.scopus.com/pages/publications/85172993439?](https://www.scopus.com/pages/publications/85172993439?origin=scopusAI)  
origin=scopusAI

128. Working with Nature for more sustainable port design; Initial findings from a 'living seawall' creation in the Port of Gladstone Mulloy, R., Aiken, C.M., Dwane, G., Jackson, E.L. Australasian Coasts and Ports 2023 Conference, 2023 <https://www.scopus.com/pages/publications/85183586537?origin=scopusAI>

129. Hydrological restoration of abandoned aquaculture ponds in North Kalimantan, Indonesia Basyuni, M., Rouf, R.A., Amelia, R., (...), Wiyono, E.B. Asian Journal of Forestry, 2025  
<https://www.scopus.com/pages/publications/105031570815?origin=scopusAI>

130. Local perceptions of blue carbon ecosystem infrastructures in Panay Island, Philippines Quevedo, J.M.D., Uchiyama, Y., Kohsaka, R. Coastal Engineering Journal, 2021  
<https://www.scopus.com/pages/publications/85102045579?origin=scopusAI>